

# A 6.5 GHz, 286 $\mu$ W Three-Stage CMOS Ring Oscillator in 65 nm Low-Power CMOS with Full Post-Layout Extraction

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**Abstract**—This paper details the complete custom physical implementation, rigorous verification, and back-end parasitic extraction of a low-power, high-frequency three-stage CMOS ring oscillator designed and laid out in a 65 nm Low-Power (LP) CMOS process. Operating under a nominal 1.0 V supply rail, the active core layout is manually optimized using row-based floorplanning to occupy a compact footprint of only 73.96  $\mu\text{m}^2$ . Rather than relying on idealized, unextracted schematic-level models, this work quantifies the performance degradation imposed by back-end-of-line (BEOL) interconnect parasitics in multi-gigahertz clocking networks. Following clean Design Rule Checking (DRC) and Layout-Versus-Schematic (LVS) matching, a comprehensive resistor-capacitor (RC) network extraction was performed in Detailed Standard Parasitic Format (DSPF). Post-layout periodic steady-state (PSS) and periodic noise (pnoise) simulation matrices reveal that distributed interconnect parasitics introduce a 41.7% reduction in loop operating frequency, shifting the carrier baseline from an unextracted schematic value of 11.15 GHz down to a physical limit of 6.50 GHz. At its 6.50 GHz steady-state operating point, the extracted netlist draws an average current of only 286.3  $\mu\text{A}$ , limiting total dynamic power dissipation to 286.3  $\mu\text{W}$ . Spectral purity analysis indicates a single-sideband phase noise floor of  $-77.6$  dBc/Hz at a 1 MHz relative frequency offset, yielding a normalized Figure of Merit (FoM) of  $-159.3$  dBc/Hz. To bypass post-layout simulation non-convergence issues caused by the complex distributed netlist, a practical workaround is documented by re-engineering the external test boundaries to match a high-impedance 1 M $\Omega$  termination floor. Overall, the results of this study indicate that a fully verified, parasitic-aware post-layout framework is important for accurately synthesizing and evaluating sub-100 nm clock distribution sub-blocks.

**Index Terms**—Ring oscillator, CMOS, 65 nm, physical design, DRC, LVS, parasitic extraction, post-layout simulation, phase noise, DSPF.

## I. INTRODUCTION

Modern ultra-large-scale integration (ULSI) transceivers and advanced system-on-chip (SoC) architectures require highly area- and power-efficient clock generation blocks, such as phase-locked loops (PLLs) and clock-and-data recovery (CDR) subsystems [1], [4]. Compared to area-intensive LC-tank topologies that necessitate thick-metal analog passives,

inductorless ring oscillators offer a minimal silicon footprint, wide frequency tuning ranges, and seamless portability across standard digital CMOS nodes [3], [11]. Among scalable ring-based configurations, the three-stage inverter feedback loop remains a structural benchmark due to its robust startup conditions, straightforward matching, and multi-phase output availability [5], [10].

However, in nanometer domains like the 65 nm node utilized here, asserting system validity solely from unextracted schematic models creates design risks. As device dimensions scale down, intrinsic transistor capacitances drop sharply, causing back-end-of-line (BEOL) interconnect parasitics to dominate the net node admittance matrix [7]. At multi-gigahertz frequencies, routing tracks introduce distributed parasitic resistances ( $R_{\text{para}}$ ) and capacitances ( $C_{\text{para}}$ ) that form unwanted RC low-pass networks. These networks inject additional phase shift into the feedback path, degrading loop phase margins and pulling the physical operating frequency away from design targets. Integrating blocks based on schematic-only metrics can therefore cause timing mismatches.

This work addresses this modeling gap by executing the step-by-step physical realization of a 6.5 GHz three-stage ring oscillator and explicitly isolating the degradation penalty imposed by layout routing. The core contributions are:

- Custom Layout Optimization: A manually drawn layout grouping row-based PMOS and NMOS arrays into a compact footprint of 73.96  $\mu\text{m}^2$ .
- Pre- vs. Post-Layout Quantification: A rigorous comparative study detailing a 41.7% frequency reduction (from 11.15 GHz down to 6.50 GHz) and a 71.2% inflation in single-stage propagation delay.
- CAD Convergence Workarounds: A documented solution for periodic steady-state (PSS) convergence failures introduced by dense extracted subcircuit formatting.

## II. ANALYTICAL MODELING & CIRCUIT ARCHITECTURE

### A. Loop Dynamics and Startup Criteria

The core topology is a three-stage, single-ended static CMOS inverting ring oscillator. Each stage contains a complementary PMOS pull-up and NMOS pull-down pair configured as a digital inverter, with the third stage output fed back to the first input to form the regenerative loop. To sustain autonomous periodic oscillations, the closed-loop system must satisfy the Barkhausen stability criteria at the oscillation frequency ( $\omega_0$ ) [1]. Modeling each stage to first order as an amplifier with a low-frequency gain of  $-\mathcal{A}_0$  and a dominant pole governed by node resistance ( $R_{\text{node}}$ ) and capacitance ( $C_{\text{node}}$ ), the single-stage transfer function is:

$$\mathcal{H}(s) = \frac{-\mathcal{A}_0}{1 + \frac{s}{\omega_p}} \quad (1)$$

where  $\omega_p = 1/(R_{\text{node}}C_{\text{node}})$  represents the  $-3$  dB open-loop pole frequency. The total loop transfer function  $\mathcal{H}_{\text{loop}}(s)$  for an  $N$ -stage configuration is:

$$\mathcal{H}_{\text{loop}}(s) = \left( \frac{-\mathcal{A}_0}{1 + \frac{s}{\omega_p}} \right)^N = \frac{(-1)^N \mathcal{A}_0^N}{\left(1 + \frac{s}{\omega_p}\right)^N} \quad (2)$$

For a three-stage loop ( $N = 3$ ), the DC negative feedback contributes an explicit phase shift of  $\pi$  ( $180^\circ$ ). To achieve the total loop phase shift of  $2\pi$  ( $360^\circ$ ) necessary for regenerative instability, the three frequency-dependent poles must collectively contribute an additional phase shift of  $\pi$ . Assuming a symmetric design, each stage contributes an equal fraction at  $\omega_0$  [1]:

$$\angle \mathcal{H}(j\omega_0) = \tan^{-1} \left( \frac{\omega_0}{\omega_p} \right) = \frac{\pi}{N} = \frac{\pi}{3} = 60^\circ \quad (3)$$

Evaluating (3) defines the exact relationship between the oscillation frequency  $\omega_0$  and the single-stage pole frequency  $\omega_p$ :

$$\omega_0 = \omega_p \tan \left( \frac{\pi}{3} \right) = \sqrt{3} \omega_p = \frac{\sqrt{3}}{R_{\text{node}} C_{\text{node}}} \quad (4)$$

Simultaneously, the loop gain magnitude must equal or exceed unity at  $\omega_0$  to guarantee that initial thermal noise undergoes exponential amplification during startup [3]. Substituting (4) back into the magnitude expression defines the minimum baseline small-signal gain required per stage:

$$\frac{\mathcal{A}_0^3}{\left[ \sqrt{1 + (\omega_0/\omega_p)^2} \right]^3} \geq 1 \implies \mathcal{A}_0 \geq \sqrt{1 + (\sqrt{3})^2} = 2 \quad (5)$$

Thus, an individual small-signal voltage gain of  $\mathcal{A}_0 \geq 2$  is mathematically mandatory to sustain stable, periodic oscillations.

### B. Short-Channel Propagation Delay Derivation

While small-signal parameters govern the onset of oscillations, the large-signal steady-state frequency is limited by the time-domain propagation delays of the inverters acting as non-linear switches [2]. The fundamental period ( $T$ ) of an

$N$ -stage ring equals the total time required for a transitioning signal edge to traverse the entire loop twice:

$$T = \frac{1}{f_{\text{osc}}} = 2Nt_d \quad (6)$$

where  $t_d$  is the single-stage propagation delay. In classical long-channel design,  $t_d$  is approximated by assuming a constant average driving current charging or discharging a single lumped capacitance across a voltage swing  $\Delta V \approx V_{\text{DD}}$ :

$$t_d \approx \frac{C_{\text{node}} V_{\text{DD}}}{2I_{\text{drive}}} \quad (7)$$

However, in deep-submicron regimes like the 65 nm Low-Power (LP) CMOS process node used here, long-channel models yield highly inaccurate estimates. Intense lateral electric fields across the ultra-short nominal gate length ( $L = 60$  nm) cause carrier velocity saturation ( $\nu_{\text{sat}}$ ) and mobility degradation to dominate device profiles [1]. Under high drain-source fields, the saturated drain current ( $I_{\text{sat}}$ ) assumes a nearly linear dependence on the gate overdrive voltage, modeled as:

$$I_{\text{sat}} = \nu_{\text{sat}} C_{\text{ox}} W (V_{\text{GS}} - V_{\text{th}} - V_{\text{dsat}}) \quad (8)$$

where  $C_{\text{ox}}$  is the gate oxide capacitance per unit area,  $W$  is the effective transistor channel width,  $V_{\text{GS}}$  is the gate-source voltage,  $V_{\text{th}}$  is the threshold voltage, and  $V_{\text{dsat}}$  is the drain saturation voltage drop.

Accounting for short-channel current restrictions during a rail-to-rail switching transition, the propagation delay  $t_d$  is formulated by integrating the time-dependent current-source constraints during the high-to-low ( $t_{\text{pHL}}$ ) and low-to-high ( $t_{\text{pLH}}$ ) transitions:

$$t_d = \frac{t_{\text{pHL}} + t_{\text{pLH}}}{2} \approx \frac{C_{\text{node}} V_{\text{DD}}}{2\nu_{\text{sat}} C_{\text{ox}} \left[ \left( \frac{W_n + W_p}{2} \right) (V_{\text{DD}} - V_{\text{th}} - V_{\text{dsat}}) \right]} \quad (9)$$

Equation (9) mathematically highlights that the physical operating frequency is strictly limited by the localized node capacitance  $C_{\text{node}}$ . In an unextracted schematic-level model,  $C_{\text{node}}$  is exclusively composed of the intrinsic device capacitances ( $C_{\text{intrinsic}} = C_{\text{gg}} + C_{\text{db}}$ ), representing the gate capacitance of the subsequent stage and the drain-bulk depletion capacitance of the driving stage.

### C. Idealized Pre-Layout Performance Baseline

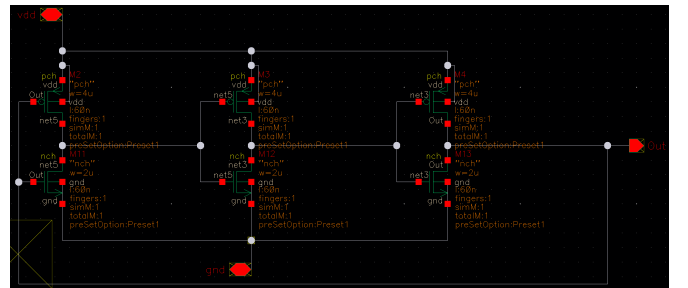


Fig. 1. Schematic capture of the verified three-stage static CMOS ring oscillator loop.

To establish a baseline for investigating physical routing penalties, the proposed three-stage ring oscillator was initially

simulated at the schematic level using a nominal supply voltage of  $V_{DD} = 1.0$  V under typical-typical (TT) corners at  $27^\circ\text{C}$ . Transistor structures utilize a uniform channel length of  $L = 60$  nm for all devices, with optimized channel widths of  $W_n = 2\ \mu\text{m}$  for the NMOS devices and  $W_p = 4\ \mu\text{m}$  for the PMOS devices to balance current driving capabilities.

TABLE I  
TRANSISTOR DIMENSIONS OF THE THREE-STAGE RING OSCILLATOR CORE

| Device Class     | Model Type | Width ( $W$ )      | Length ( $L$ ) |
|------------------|------------|--------------------|----------------|
| NMOS (3 devices) | nch        | $2.0\ \mu\text{m}$ | 60 nm          |
| PMOS (3 devices) | pch        | $4.0\ \mu\text{m}$ | 60 nm          |

Under these clean, layout-parasitic-free conditions, the intrinsic charging and discharging time constants are brief. The pre-layout transient simulation yields a high oscillation frequency of  $f_{\text{osc,schematic}} = 11.15$  GHz, corresponding to an ultra-short total loop period of  $T_{\text{schematic}} = 89.71$  ps. Applying (6), the implied large-signal propagation delay per individual inverter stage evaluates to  $t_{d,\text{schematic}} = T_{\text{schematic}}/2N \approx 14.95$  ps. This value represents the physical performance ceiling of the chosen sizes, assuming ideal zero-length inter-stage connections.

### III. PHYSICAL DESIGN & CUSTOM LAYOUT METHODOLOGY

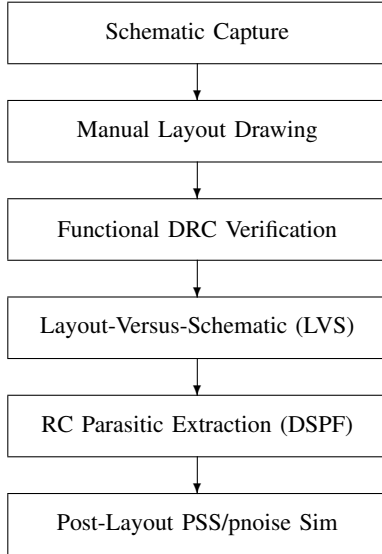


Fig. 2. Custom physical-design and parasitic-aware verification flow framework implemented in this work.

#### A. Device Sizing and Symmetrical Matching Strategy

Symmetrical driving current capability was enforced in the TSMC 65 nm LP process to ensure balanced rail-to-rail voltage swings. Large-signal delay components ( $t_{pHL}$  and  $t_{pLH}$ ) depend heavily on the transistor on-resistance ( $R_{on}$ ) and total output

loading capacitance [2]. Because hole mobility ( $\mu_p$ ) is fundamentally lower than electron mobility ( $\mu_n$ ), a geometric sizing ratio of 2 : 1 was locked ( $W_p = 4\ \mu\text{m}$ ,  $W_n = 2\ \mu\text{m}$ ). Drawn physical gate lengths are kept at the minimum  $L = 60$  nm to maximize current density, minimize intrinsic gate charge ( $Q_g$ ), and achieve short channel transit times [1]. This symmetrical design balances switching transitions, suppressing systematic duty-cycle distortion (DCD) and reducing up-converted  $1/f$  flicker noise contributions.

#### B. Floorplanning and Interconnect Routing Topology

The core was realized through a fully manual custom layout to preserve high-frequency signal integrity. The topology follows an efficient two-row floorplanning architecture: the three PMOS devices are aligned longitudinally in an upper row within a single, shared n-well island, while the three NMOS devices are mirrored in the lower row within the p-substrate. Grouping the PMOS devices eliminates separate well boundary overheads and minimizes foundry spacing rules. To suppress BEOL routing parasitics, all inter-stage loop feedback nets were routed using short, wide tracks in the lower metallization layers (Metal 1 and Metal 2) [2]. This bounding strategy limits via stack heights, preventing excess parasitic resistance from dampening loop gains. The resulting physical oscillator core footprint is  $8.6\ \mu\text{m} \times 8.6\ \mu\text{m}$ , yielding a total core area of  $73.96\ \mu\text{m}^2$ .

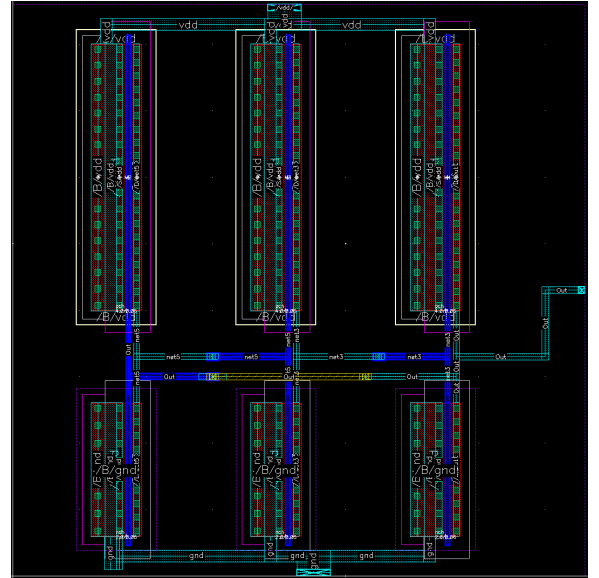


Fig. 3. Manually routed layout core showing row-aligned PMOS and NMOS device matching ( $8.6\ \mu\text{m} \times 8.6\ \mu\text{m}$ ).

#### C. Physical Verification: Design Rule Checking (DRC)

The manual cell layout underwent strict automated Design Rule Checking (DRC) to verify compliance with the sub-micron lithographic rules of the 65 nm process node. An initial Metal-1 minimum spacing violation (rule M1.S.1, min  $0.09\ \mu\text{m}$ ) discovered along the inter-stage interconnect paths was resolved by re-adjusting track separations without expanding cell

boundaries. The refined layout achieved complete functional DRC cleanliness. Remaining open flags reflect standard foundry layer density alerts and chip-level seal-ring rules (CSR and AP layers). These informational items do not impact electrical performance and are routinely addressed via automated dummy feature insertion during final mask assembly.

#### D. Layout-Versus-Schematic (LVS) Verification

To guarantee that the physical geometries drawn in the layout editor map directly onto the intended loop topology, a comprehensive Layout-Versus-Schematic (LVS) check was executed. The LVS tool cross-examined extracted device types, terminal pins, net connections, and transistor component dimensions ( $W/L$ ) against the source schematic. LVS verification yielded a perfect match, confirming the total absence of shorts, opens, or device mismatches. This clean baseline confirms that all subsequent performance deviations observed during final characterization stem entirely from interconnect physics rather than structural coding discrepancies.

TABLE II  
EXTRACTED PER-NET PARASITIC CAPACITANCE MATRIX FROM DSPF

| Extracted Net Label | Parasitic Capacitance (fF) |
|---------------------|----------------------------|
| Out                 | 4.07 fF                    |
| net3                | 3.54 fF                    |
| net5                | 3.52 fF                    |
| Vdd                 | 3.16 fF                    |
| gnd                 | 3.01 fF                    |

### IV. PARASITIC-AWARE EXTRACTION & SIMULATION CONVERGENCE CHALLENGES

#### A. Resistor-Capacitor (RC) Netlist Extraction Matrix

Following physical verification, back-end parasitic extraction was executed to generate a simulation-ready network. The extraction engine analyzed active interconnect layers, contact vias, and substrate boundaries to create an exhaustive transistor-level netlist annotated with distributed RC elements in Detailed Standard Parasitic Format (DSPF) [2].

This process isolates the parasitic loading introduced by manual routing across the three loop nodes. As detailed in Table II, the active output node (Out) exhibits the highest loading at 4.07 fF, driven by the added trace length required for the loop feedback path and the external testing port. Internal nodes *net3* and *net5* display symmetrical parasitic capacitance baselines of 3.54 fF and 3.52 fF. Additionally, the localized power rails (Vdd and gnd) carry 3.16 fF and 3.01 fF of parasitic capacitance to the substrate. Together, these femtofarad-scale capacitances increase the node loading, introducing a low-pass effect within the signal loop.

#### B. The Periodic Steady-State (PSS) Convergence Solution

Simulating post-layout, RC-extracted oscillator netlists using Periodic Steady-State (PSS) and periodic noise (*pnoise*) shooting algorithms presents severe numerical convergence

challenges. Initial configurations failed repeatedly, logging non-convergence errors due to a boundary condition loading mismatch at the output node. By default, the standard *analogLib* simulation port primitive introduces a  $50\ \Omega$  source/load resistance. While a  $50\ \Omega$  termination matches standard test instruments, it imposes an extreme load when connected directly to an unbuffered, low-power sub-micron CMOS inverter stage. This load severely damped active loop gain below the Barkhausen criterion ( $\mathcal{A}_0 < 2$ ), causing an anomalous supply current draw of 19.59 mA while extinguishing the oscillation entirely.

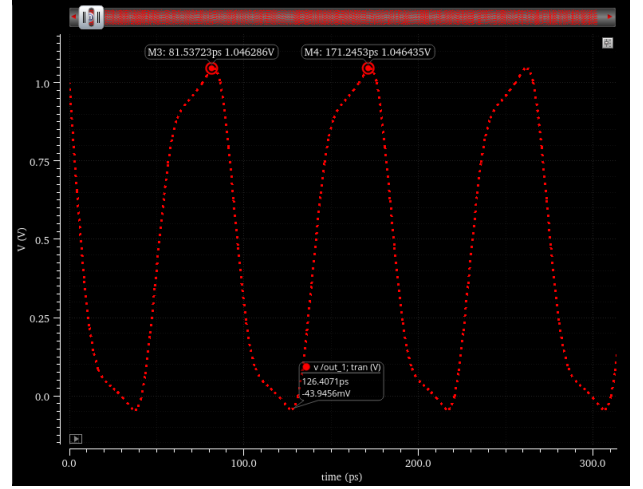


Fig. 4. Idealized pre-layout (schematic-level) transient output waveform demonstrating an unextracted oscillation frequency of 11.15 GHz ( $T = 89.71$  ps).

To resolve this limitation and force PSS convergence on the dense extracted netlist, the simulation parameters were re-engineered. The internal resistance of the output port was increased from  $50\ \Omega$  to a high-impedance threshold of  $1\ \text{M}\Omega$ , successfully removing the AC loading penalty from the feedback loop. Symmetrical engine variables were optimized as follows:

- **Oscillator Mode Activation:** Configured with the feedback node set explicitly to the loop output (*node+ = out\_1*), matching the target beat frequency directly to the anticipated baseline.
- **Enforced Initial Conditions (IC):** An explicit voltage condition of  $\text{IC} = 1.0\ \text{V}$  was applied to the internal node to trigger immediate regenerative startup.
- **Stabilization Interval ( $t_{\text{stab}}$ ):** Extended to  $t_{\text{stab}} = 50\ \text{ns}$  to force the transient analysis to completely settle into a stable limit cycle before calculating the shooting matrix.
- **Tolerance Preset:** Locked to *moderate* to balance numerical stability with multi-gigahertz Fourier matrix resolution.

For the subsequent *pnoise* analysis, the sweep range spanned 1 kHz to 100 MHz relative to the moving carrier. The noise type was bound to *timeaverage* with noise contributions limited to phase modulation (PM), and the output probe locked directly onto the high-impedance external port (*oprobe = port*). This convergence framework eliminated

anomalous current spikes, enabling stable extraction modeling without requiring any layout parasitic simplifications.

## V. POST-LAYOUT SIMULATION RESULTS

### A. Post-Layout Transient Performance Profile

Following PSS numerical stabilization, the post-layout transient behavior of the RC-extracted oscillator netlist was fully characterized under a nominal 1.0 V supply rail. Fig. 5 illustrates the post-layout transient waveform, which exhibits stable rail-to-rail periodic oscillations with a voltage swing spanning  $-0.05$  V up to  $1.04$  V.

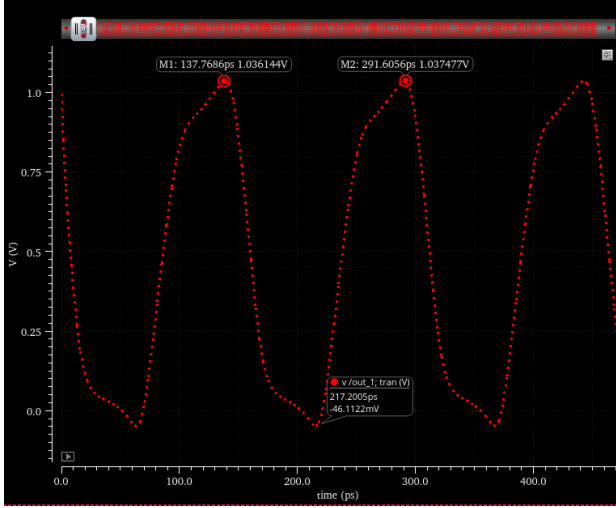


Fig. 5. Post-layout (RC-extracted) transient output waveform showing stable rail-to-rail oscillation at 6.50 GHz with a period of 153.84 ps.

By placing adjacent peak markers ( $M_1$  and  $M_2$ ) on the settled time grid, the physical oscillation period was measured at  $t_1 = 137.77$  ps and  $t_2 = 291.61$  ps, yielding an exact period of:

$$T_{\text{post-layout}} = t_2 - t_1 = 153.84 \text{ ps} \quad (10)$$

The reciprocal of this period defines the true physical operating frequency of the realized oscillator core:

$$f_{\text{osc,post-layout}} = \frac{1}{T_{\text{post-layout}}} = 6.50 \text{ GHz} \quad (11)$$

The dynamic supply current monitored over a full cycle via the expression `average(IT("V0/PLUS"))` confirms a stable current draw of  $|I_{\text{avg}}| = 286.3 \mu\text{A}$ . The total dynamic power consumption ( $P_{\text{dc}}$ ) evaluates to:

$$P_{\text{dc}} = V_{\text{DD}} \times |I_{\text{avg}}| = 1.0 \text{ V} \times 286.3 \mu\text{A} = 286.3 \mu\text{W} \quad (12)$$

Using (6), the true large-signal propagation delay per individual inverter stage expands to  $t_{\text{d,post-layout}} = T/(2N) = 25.6$  ps.

### B. Pre- vs. Post-Layout Performance Degradation Analysis

Direct comparisons isolate the performance penalties imposed by physical interconnect routing. The operating frequency drops from an unextracted schematic baseline of 11.15 GHz down to a physical limit of 6.50 GHz, a 41.7% frequency reduction. Because transistor aspects ( $W/L$ ) and internal current baselines remained perfectly identical

across both setups, this delay inflation is driven entirely by the distributed RC parasitics embedded in the DSPF netlist. This parasitic load inflates the large-signal stage delay from 14.95 ps up to 25.6 ps, representing a 71.2% inflation in stage delay. This divergence underscores that parasitic extraction is mandatory to accurately predict timing parameters in multi-gigahertz nanometer layouts.

### C. Spectral Purity and Figure of Merit Evaluation

The phase noise spectrum of the realized oscillator was characterized using the converged high-impedance `pnoise` simulation. As shown in Fig. 6, the noise curve exhibits a consistent  $1/f^2$  slope across the offset spectrum, confirming that white thermal noise contributions dominate the high-frequency phase error profile [6], [9].

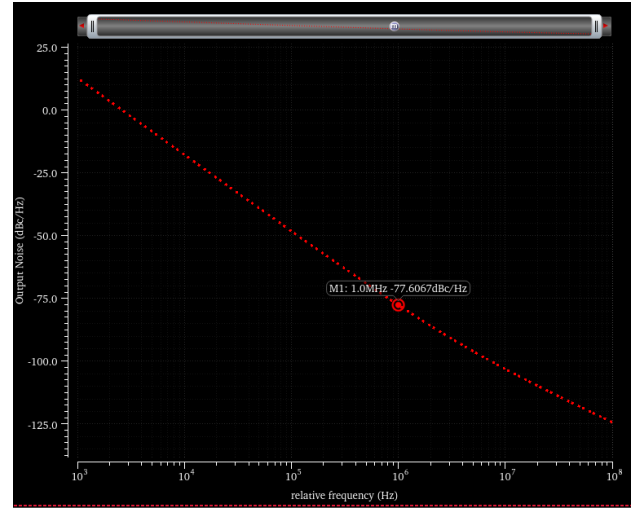


Fig. 6. Post-layout (RC-extracted) single-sideband phase noise versus relative offset frequency, showing  $-77.6$  dBc/Hz at a 1 MHz offset.

At a relative offset of  $\Delta f = 1$  MHz away from the 6.50 GHz carrier, the oscillator achieves a phase noise baseline of  $-77.6$  dBc/Hz. At an offset of  $\Delta f = 100$  MHz, the noise floor drops cleanly to  $-124.3$  dBc/Hz. To evaluate performance balancing across frequency, power, and spectral purity, metrics were normalized using the standard IEEE oscillator Figure of Merit (FoM) formulation [8]:

$$\text{FoM} = \mathcal{L}(\Delta f) - 20 \log_{10} \left( \frac{f_{\text{osc}}}{\Delta f} \right) + 10 \log_{10} \left( \frac{P_{\text{dc}}}{1 \text{ mW}} \right) \quad (13)$$

where  $\mathcal{L}(\Delta f)$  represents the single-sideband phase noise. Substituting the extracted post-layout metrics ( $\mathcal{L}(1 \text{ MHz}) = -77.6$  dBc/Hz,  $f_{\text{osc}} = 6.50$  GHz, and  $P_{\text{dc}} = 286.3 \mu\text{W}$ ) into (13) provides the following evaluation:

$$\begin{aligned} \text{FoM} &= -77.6 - 20 \log_{10} \left( \frac{6.5 \times 10^9}{1.0 \times 10^6} \right) + 10 \log_{10} (0.2863) \\ &= -77.6 - 76.26 - 5.43 = -159.3 \text{ dBc/Hz} \end{aligned} \quad (14)$$

An FoM of  $-159.3$  dBc/Hz confirms that the compact, manually optimized physical design maintains an efficient balance between power consumption and spectral purity within the 65 nm node. It should be noted that this favorable FoM is aided



by the sub-milliwatt power consumption; the absolute phase noise of  $-77.6$  dBc/Hz at 1 MHz is typical of an unoptimized single-ended ring oscillator and leaves room for improvement through established low-noise delay-cell techniques.

#### D. Qualitative Context with Prior Art

A direct quantitative comparison with prior ring oscillators is not attempted here, since most reported designs are measured, frequency-tunable oscillators, whereas this work characterizes a fixed-frequency core through post-layout simulation. Reported CMOS ring oscillators span a wide range of technology nodes and tuning strategies, including current-controlled and quadrature topologies in  $0.18\text{ }\mu\text{m}$  CMOS [10], [11], injection-based designs for all-digital PLLs [12], and current-starved ring VCOs targeting 5G bands [13]. Relative to these, the present design is distinguished less by raw performance than by the fully verified, parasitic-aware methodology used to obtain its figures; the compact  $73.96\text{ }\mu\text{m}^2$  core and  $286.3\text{ }\mu\text{W}$  power consumption are reported as simulated post-layout values rather than measured silicon.

### VI. CONCLUSION

This paper presented the custom physical design, layout realization, and post-layout verification of a low-power three-stage CMOS ring oscillator in a 65 nm LP CMOS baseline. By executing a manual cell floorplanning strategy, the core layout area was optimized down to a compact footprint of  $73.96\text{ }\mu\text{m}^2$ . Under a nominal 1.0 V supply rail, the fully extracted RC netlist achieved stable physical oscillations at 6.50 GHz while restricting core power consumption to  $286.3\text{ }\mu\text{W}$ . Periodic noise simulations verified a single-sideband phase noise baseline of  $-77.6$  dBc/Hz at a 1 MHz offset from the carrier, evaluating to a competitive normalized FoM of  $-159.3$  dBc/Hz.

A pre- and post-layout comparison quantified a 41.7% frequency reduction from the idealized schematic baseline of 11.15 GHz, illustrating how strongly BEOL routing parasitics degrade deep-submicron clock trees. This performance shift corresponds to a 71.2% inflation in the single-stage propagation delay (increasing from 14.95 ps up to 25.6 ps), indicating that schematic-level estimates can substantially overstate achievable performance in nanometer regimes. To resolve periodic steady-state (PSS) CAD convergence failures introduced by the dense extracted netlist, the output boundary condition was re-engineered by matching the external test port to a high-impedance  $1\text{ M}\Omega$  floor. Overall, the results indicate that a fully verified, parasitic-aware implementation framework is important for synthesizing multi-gigahertz clock generation blocks in modern system-on-chip architectures.

The present results are based on post-layout, parasitic-extracted simulation; silicon measurement is left as future work. Future extensions also include adding explicit frequency tuning and applying low-noise delay-cell techniques to improve the absolute phase noise.

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